RAG Learning

C1 — Air and Water

OCR GCSE Chemistry B (J258) — Twenty First Century Science · All papers (J258/01 & J258/03 Breadth in Chemistry, J258/02 & J258/04 Depth in Chemistry)

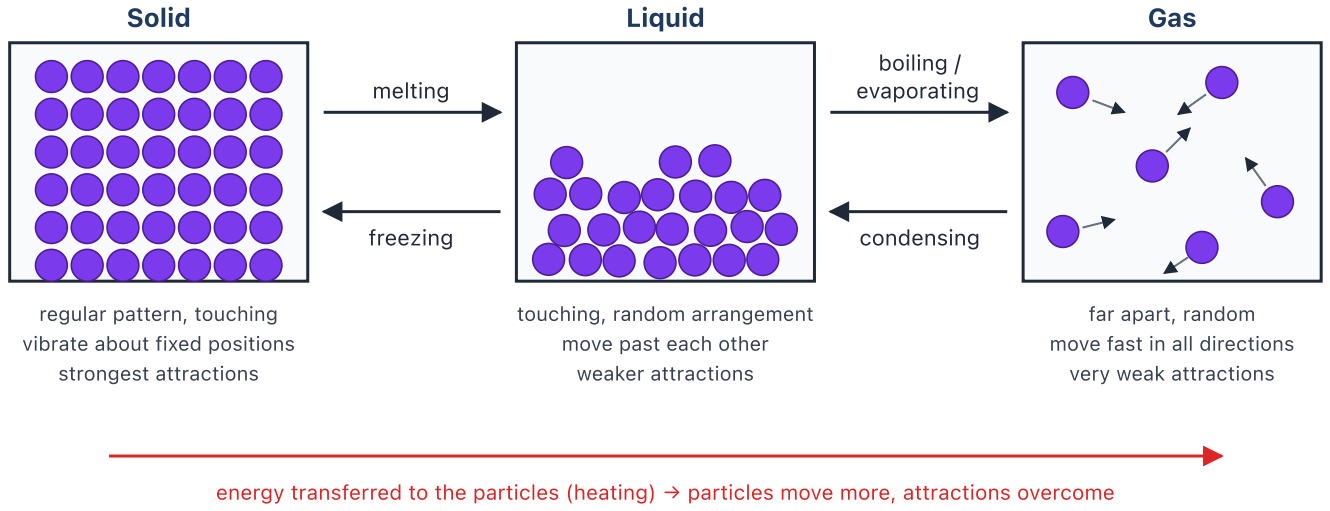
Content marked **(HT)** is Higher tier only (papers J258/03 and J258/04). Everything else can be examined on both tiers.

C1.1 How Has the Earth's Atmosphere Changed Over Time, and Why?

The Particle Model

> The **particle model** says all matter is made of tiny particles (atoms, molecules or ions) that are constantly moving and that attract one another. The **arrangement**, **movement** and **strength of attraction** between the particles explain the three states of matter.

The Particle Model: States and Changes of State



State	Arrangement	Movement	Bulk properties explained
Solid	Close together, regular pattern	Vibrate about fixed positions	Fixed shape and volume; cannot be compressed
Liquid	Close together (touching), random	Slide past each other	Flows, takes the shape of the container; fixed volume; hardly compressible
Gas	Far apart, random	Fast, random, in all directions	Fills the container; easily compressed; particles hitting the walls cause gas pressure

Physical and Chemical Changes

- A **physical change** (e.g. melting, boiling, dissolving) does *not* make a new substance. The particles themselves are unchanged — only their arrangement and movement change — so it is easily reversed and mass is conserved.
- A **chemical change** (a reaction, e.g. burning) makes **new substances**: atoms are rearranged and bonds are broken and made. It is usually hard to reverse. Mass is still conserved.

> **Exam Tip:** The particles do **not** have the properties of the bulk substance. A single copper atom is not orange and does not conduct; a sulfur atom is not yellow; a single water molecule is not "wet" and does not "melt". Particles do not expand when heated — the *spaces* between them get bigger. Writing "the particles melt" or "the particles get bigger" loses marks.

Limitations of the Particle Model (HT)

Diagrams of particles as small solid spheres are a useful **model**, but they have limitations (HT):

- They do not show the **scale** correctly — real particles are far smaller, and in a gas the spaces between particles are much larger than can be drawn (about 10 particle diameters apart in air).
- They treat particles as **inelastic (hard, solid) spheres**. Real atoms and molecules are not hard balls: they have a nucleus and electrons, can have different shapes, and can deform or vibrate internally.

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- They do not fully show the **forces between particles** — the different strengths of attraction in different substances, and the fact that particles attract over a distance, not only when touching.
- They do not show particle **motion** (speeds vary and change continually) in a still picture.

Why Substances Change State at Different Temperatures

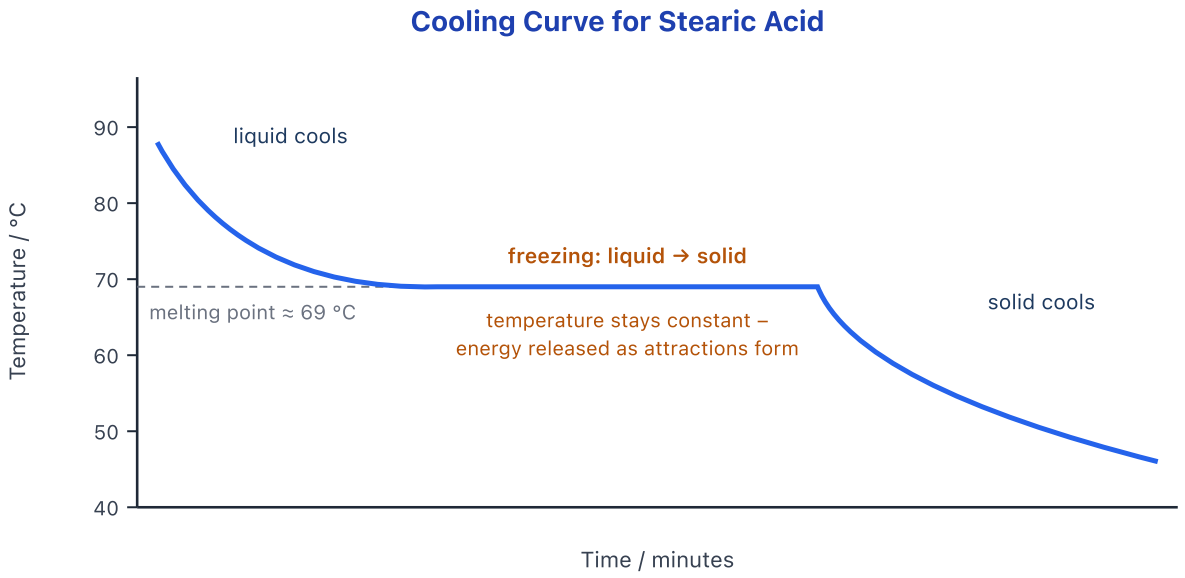
To melt or boil a substance, **energy must be transferred to it** (by heating) to overcome the forces of attraction between its particles. When a liquid freezes or a gas condenses, attractions form and energy is transferred to the surroundings.

- **Weak forces** between particles (e.g. between small molecules such as O_2, N_2, CH_4) → little energy needed → **low** melting and boiling points (gases at room temperature).
- **Strong forces** (e.g. between ions in sodium chloride or atoms in a metal) → lots of energy needed → **high** melting and boiling points.
- During a change of state the **temperature stays constant**: the energy transferred is used to break (or is released by forming) attractions between particles, not to change the speed of the particles.

Practical: Cooling Curve for Stearic Acid (or Heating Curve for Ice)

Melt stearic acid in a boiling tube in a hot water bath, put a thermometer in it, remove it from the water and record the temperature every 30 seconds as it cools. Plot temperature (y-axis) against time (x-axis). The flat section shows the freezing (melting) point.

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Predicting States from Data

Compare the temperature with the substance's melting point (mp) and boiling point (bp):

- below mp → **solid**
- between mp and bp → **liquid**
- above bp → **gas**

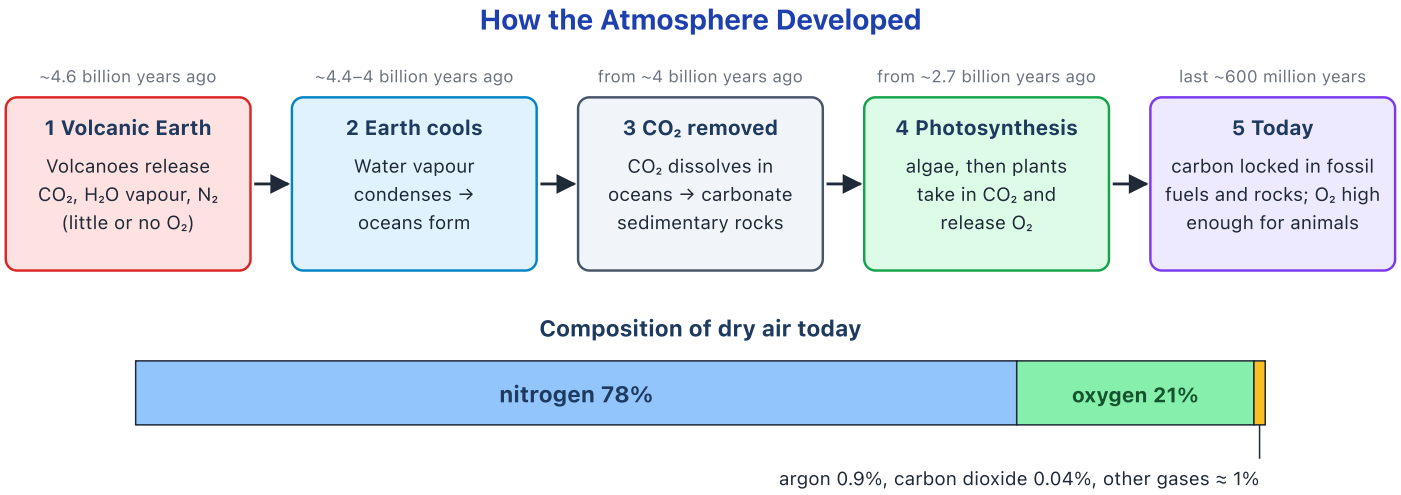
Substance	Melting point / °C	Boiling point / °C	State at 20 °C
Nitrogen, N_2	−210	−196	gas
Water, H_2O	0	100	liquid
Bromine, Br_2	−7	59	liquid
Stearic acid	69	361	solid
Sodium chloride, NaCl	801	1413	solid

Worked Example: What state is bromine in at −20 °C, and at 80 °C?
−20 °C is below its mp (−7 °C) → **solid**. 80 °C is above its bp (59 °C) → **gas**.

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How the Atmosphere Formed and Changed

Scientists' explanations are based on **evidence**, including the types and chemical composition of **ancient rocks** (e.g. iron-rich rock layers that could only form once oxygen was present) and **fossils** of early life (e.g. stromatolites made by photosynthesising bacteria).



1. **Early volcanic activity** released gases — mainly **carbon dioxide** and **water vapour**, with nitrogen and small amounts of other gases (e.g. methane, ammonia). There was little or no oxygen.
2. The Earth **cooled**; water vapour **condensed** (a change of state: gas → liquid) and formed the **oceans**.
3. Carbon dioxide **dissolved in the oceans** and was locked up in **carbonate sedimentary rocks** (e.g. limestone, from shells and skeletons of marine organisms).
4. **Photosynthesising organisms** (algae, then plants) evolved: carbon dioxide + water → glucose + oxygen. They removed CO_2 and slowly built up **oxygen**.
5. Carbon from dead organisms was trapped as **fossil fuels** (coal, oil, gas). The rising oxygen level allowed **animals** to evolve.

Today's dry air: about 78% nitrogen, 21% oxygen, 0.9% argon, 0.04% carbon dioxide, plus traces of other gases. The amount of water vapour varies.

> **Ideas about Science (IaS3) — data vs explanations.** *Data* are the observations (e.g. "rocks older than 2.4 billion years contain unoxidised iron minerals"). An *explanation* is the idea proposed to account for them (e.g. "the early atmosphere contained almost no

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oxygen"). Nobody observed the early atmosphere, so explanations are developed creatively from evidence and can change as new evidence (e.g. new rock samples or fossils) is found. The particle model is also used here to explain state changes such as the formation of the oceans and the water cycle (evaporation, condensation).

Air Pollutants from Burning Fuels

Our demand for energy means large amounts of fossil fuels are burned for transport and electricity generation. Combustion produces pollutants:

Pollutant	Major source	Problems caused
Carbon monoxide, CO	**Incomplete combustion** of fuels (not enough oxygen)	Toxic: combines with haemoglobin so blood carries less oxygen. Colourless and odourless, so hard to detect.
Particulates (soot, mostly carbon)	**Incomplete combustion**, especially in diesel engines	Breathed deep into the lungs → respiratory and heart problems; blacken buildings; contribute to **smog** and global dimming.
Sulfur dioxide, SO_2	Combustion of **sulfur impurities** in fuels (coal, petrol, diesel)	Causes **acid rain** (damages trees, lakes, buildings); irritates the lungs and worsens asthma.
Nitrogen oxides, NO_x (NO, NO_2)	Nitrogen and oxygen from the **air** react at the **high temperatures** in engines and power stations; NO is then **further oxidised** to NO_2 in the air	**Acid rain**, photochemical **smog**, lung irritation and breathing problems.

Nitrogen oxides: $N_2(g) + O_2(g) \rightarrow 2NO(g)$ then $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Sulfur dioxide: $S(s) + O_2(g) \rightarrow SO_2(g)$

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Reducing Emissions

Approach	How it works
Catalytic converters (car exhausts)	A hot catalyst (platinum/rhodium on a honeycomb with a large surface area) converts CO to CO_2 and NO back to N_2: $2CO + 2NO \rightarrow 2CO_2 + N_2$. They also oxidise unburnt fuel to CO_2 and H_2O.
Low-sulfur petrol and diesel	Sulfur is removed from the fuel at the refinery, so less SO_2 is formed when it burns.
Gas scrubbers (power station chimneys)	Flue gases are passed through an alkaline slurry, e.g. calcium oxide or calcium carbonate (limestone) with water, which reacts with the acidic SO_2 and removes it.
Other measures	Particulate filters on diesel engines; using less fuel (efficient engines, public transport); switching to electric vehicles or renewable electricity.

> **Ideas about Science (IaS4) — impacts of science.** Burning fossil fuels has given us transport and electricity, but has had *unintended* impacts on air quality. Catalytic converters, low-sulfur fuels and gas scrubbers are *positive applications* of science that reduce those impacts. Scientists monitor pollutant concentrations so that air quality can be maintained and regulated.

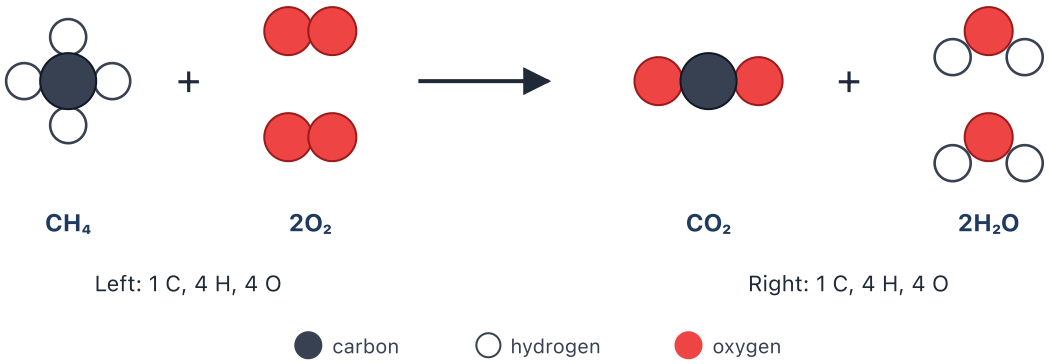
Formulae and Balanced Equations

Many non-metal elements exist as **diatomic molecules**: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2. Metals and the noble gases are written as single symbols (Fe, Na, He). Simple covalent compounds include H_2O, CO_2, CO, CH_4, NH_3, SO_2, NO, NO_2 and HCl.

Atoms are not created or destroyed in a reaction (**conservation of mass**), so a symbol equation must have the same number of each type of atom on both sides. Balance by changing the **numbers in front** of formulae — never the small subscript numbers inside a formula.

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Complete Combustion of Methane: Atoms Are Rearranged, Not Lost



Worked Example: Balance the combustion of propane: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

1. Carbon: 3 on the left, so $3CO_2$.
2. Hydrogen: 8 on the left, so $4H_2O$.
3. Oxygen on the right: $(3 \times 2) + (4 \times 1) = 10$ atoms = 5 O_2.

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ Check: C 3 = 3, H 8 = 8, O 10 = 10 ✓

Worked Example: Incomplete combustion of methane making carbon monoxide: $CH_4 + O_2 \rightarrow CO + H_2O$

Use ratios: 1 CH_4 gives 1 CO and 2 H_2O, which need 3 O atoms = 1½ O_2. Double everything to remove the fraction:

$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$

Combustion is Oxidation

Oxidation is the **gain of oxygen** by a substance. When a fuel burns, its carbon and hydrogen combine with oxygen (C → CO_2, H → H_2O), so **combustion is an oxidation reaction**. Forming SO_2 from sulfur and NO/NO_2 from nitrogen are also oxidations.

Tests for Gases

Gas	Test	Positive result

Oxygen, O_2	Insert a **glowing splint**	Splint **relights**
Hydrogen, H_2	Hold a **lighted splint** at the mouth of the tube	**Squeaky pop** (hydrogen burns: $2H_2 + O_2 \rightarrow 2H_2O$)
Carbon dioxide, CO_2	Bubble the gas through **limewater** (calcium hydroxide solution)	Limewater turns **milky / cloudy** (white precipitate of calcium carbonate)

These tests are part of **PAG2** (identifying the gases produced, e.g. in electrolysis) and the chlorine test is in C1.4.

> **Exam Tip:** Don't mix up the two splint tests: **glowing** splint → oxygen **relights** it; **lighted** splint → hydrogen gives a **squeaky pop**. "Limewater goes cloudy" is the only accepted test for CO_2 — "puts out a lighted splint" is not a test, since many gases do that.

C1.2 Why Are There Temperature Changes in Chemical Reactions?

Exothermic and Endothermic Reactions

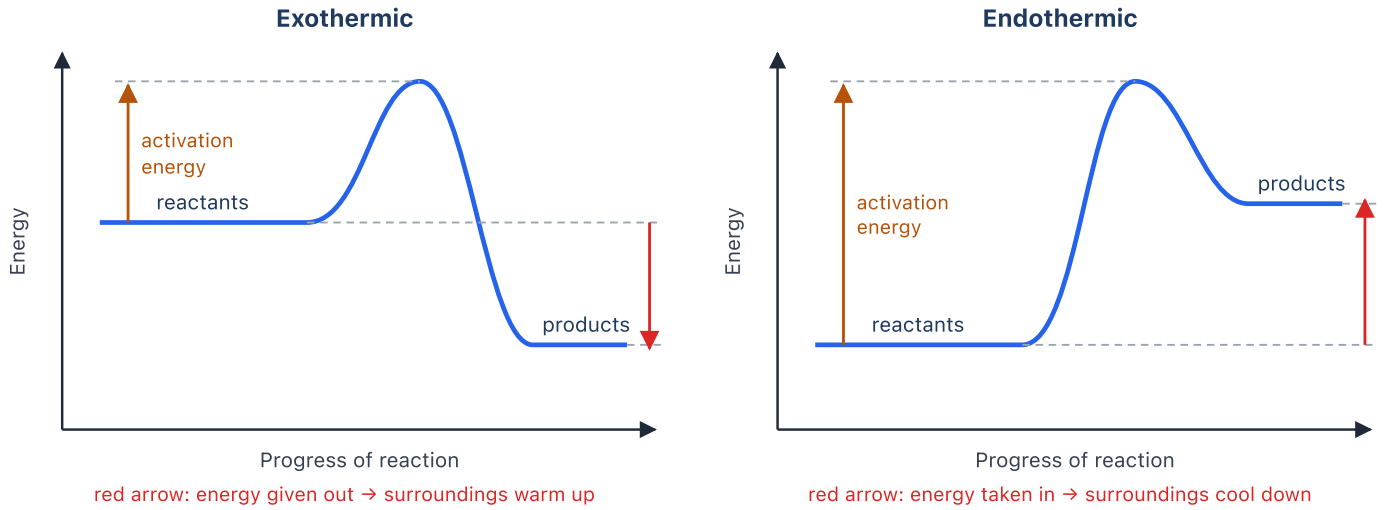
	Exothermic	Endothermic
Energy transfer	Energy transferred **to** the surroundings	Energy taken in **from** the surroundings
Temperature of surroundings	**Rises** (the mixture/thermometer gets warmer)	**Falls** (gets colder)
Examples	Combustion of fuels, neutralisation, most oxidation reactions, respiration, metals with acids	Thermal decomposition (e.g. of calcium carbonate), citric acid + sodium hydrogencarbonate, dissolving ammonium nitrate, photosynthesis

Reaction Profiles and Activation Energy

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The **activation energy** is the **minimum energy needed for a reaction to occur** — the energy needed to start breaking bonds in the reactants. This is why a fuel needs a spark or flame before it burns.

Reaction Profiles



- Activation energy = height from the **reactants** to the **top of the hump**.
- Overall energy change = difference between reactants and products. Exothermic: products **lower** than reactants. Endothermic: products **higher**.

Bond Breaking and Bond Making

- **Breaking bonds** needs energy — it is **endothermic**.
- **Making bonds** releases energy — it is **exothermic**.
- If more energy is released making new bonds than is needed to break the old ones, the reaction is **exothermic** overall. If less, it is endothermic.

Calculating Energy Changes from Bond Energies (HT)

energy change = energy needed to break bonds − energy released making bonds

negative answer = exothermic · positive answer = endothermic

Bond	H–H	Cl–Cl	H–Cl	C–H	O=O	C=O	O–H

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Bond energy / kJ mol^{-1}	436	243	432	413	498	805	464

Worked Example: (HT) Calculate the energy change for $H_2 + Cl_2 \rightarrow 2HCl$.

Bonds broken: 1 × H–H + 1 × Cl–Cl = 436 + 243 = 679 kJ

Bonds made: 2 × H–Cl = 2 × 432 = 864 kJ

Energy change = 679 – 864 = **–185 kJ/mol** → exothermic.

Worked Example: (HT) Calculate the energy change for burning methane: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$.

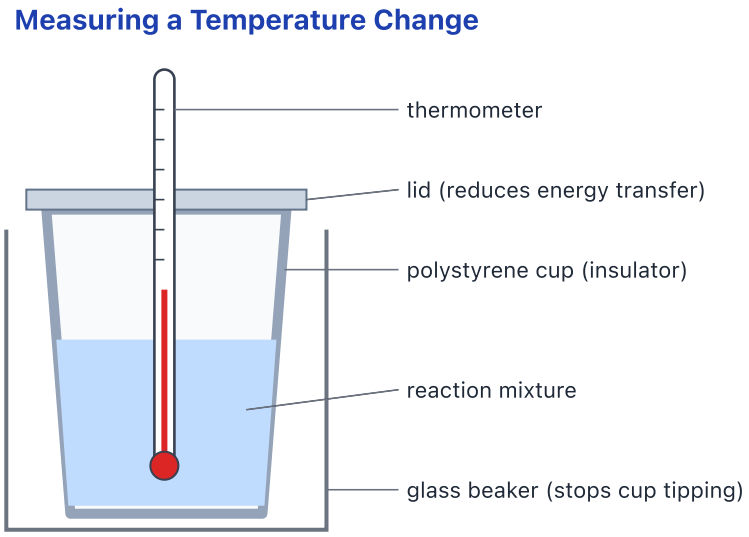
Broken: 4 × C–H + 2 × O=O = (4 × 413) + (2 × 498) = 1652 + 996 = 2648 kJ

Made: 2 × C=O + 4 × O–H = (2 × 805) + (4 × 464) = 1610 + 1856 = 3466 kJ

Energy change = 2648 – 3466 = **–818 kJ/mol** → exothermic.

Exam Tip: Draw out every molecule's bonds before counting — CO_2 has **two** C=O bonds and each H_2O has **two** O–H bonds. Multiply by the balancing numbers. Always state "exothermic" or "endothermic" from the sign.

Practical: Is a Reaction Exothermic or Endothermic?



Record start temperature, add reactant, stir, record highest/lowest temperature

1. Measure a fixed volume (e.g. 25 cm^3) of one reactant solution into a **polystyrene cup** standing in a beaker (for stability).
2. Record the starting temperature with a thermometer (or temperature probe).
3. Add the second reactant (a measured mass or volume), put on the **lid**, and stir.
4. Record the highest (or lowest) temperature reached.
5. Temperature rise → exothermic; temperature fall → endothermic. Repeat to check the results are repeatable.

The polystyrene cup and lid are **insulators**: they reduce energy transfer to or from the surroundings so the temperature change measured is closer to the true value.

Chemical Cells and Fuel Cells

A **chemical cell** (e.g. a battery) uses a chemical reaction to produce a **potential difference**. It keeps producing a potential difference **until the reactants are used up**; then it is "flat". Non-rechargeable cells must then be replaced.

A **fuel cell** is supplied continuously with a fuel and oxygen, so it keeps producing a potential difference as long as the fuel is supplied. In a hydrogen–oxygen fuel cell the overall reaction is the same as burning hydrogen, and the only product is water — but the energy is transferred electrically to drive an electric motor instead of heating gases in an internal combustion engine.

$$\text{hydrogen} + \text{oxygen} \rightarrow \text{water} \qquad 2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Advantages of hydrogen fuel cells (for vehicles)	Disadvantages

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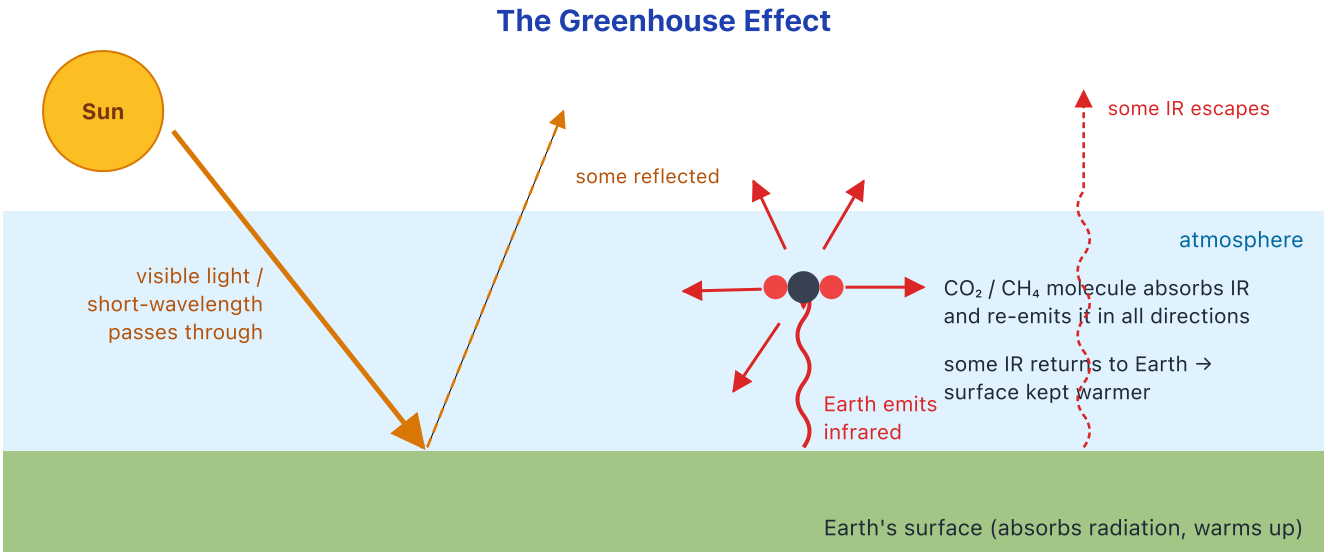
• Only product is water — no CO_2, CO, NO_x or particulates at the point of use, improving air quality in cities • More efficient than internal combustion engines; fewer moving parts • Refuel quickly; longer range than many battery cars	• Hydrogen is usually made by electrolysis (or from natural gas); if the electricity comes from fossil fuels, pollutants are produced *elsewhere* • Hydrogen is a gas: hard to store (high-pressure tanks), flammable/explosive • Few refuelling stations; fuel cells are expensive (they use platinum catalysts)

Other fuel cells (e.g. methanol fuel cells) are easier to refuel because the fuel is a liquid, but they produce CO_2. Rechargeable batteries are an alternative for cars and phones but take longer to recharge and contain toxic metal compounds that must be disposed of carefully.

> **Ideas about Science (IaS4).** Fuel cells are a *positive application of science* that can reduce emissions where people live. When evaluating a technology, consider the whole picture: where the hydrogen comes from, the cost, the risks of storage, and who benefits.

C1.3 What Is the Evidence for Climate Change, and Why Is It Occurring?

The Greenhouse Effect



More CO_2 and CH_4 → more infrared absorbed and re-emitted → higher average temperature (global warming)

1. Electromagnetic radiation from the Sun (mostly visible light, shorter wavelength) passes through the atmosphere and is **absorbed** by the Earth's surface, warming it. Some is reflected.
2. The warm Earth **emits infrared radiation** (longer wavelength).
3. **Greenhouse gases** — e.g. **carbon dioxide, methane** and water vapour — **absorb** some of this infrared and **re-emit** it **in all directions**, including back towards the Earth.
4. This keeps the Earth warmer than it would otherwise be. Without any greenhouse effect the average surface temperature would be about –18 °C — too cold for most life.

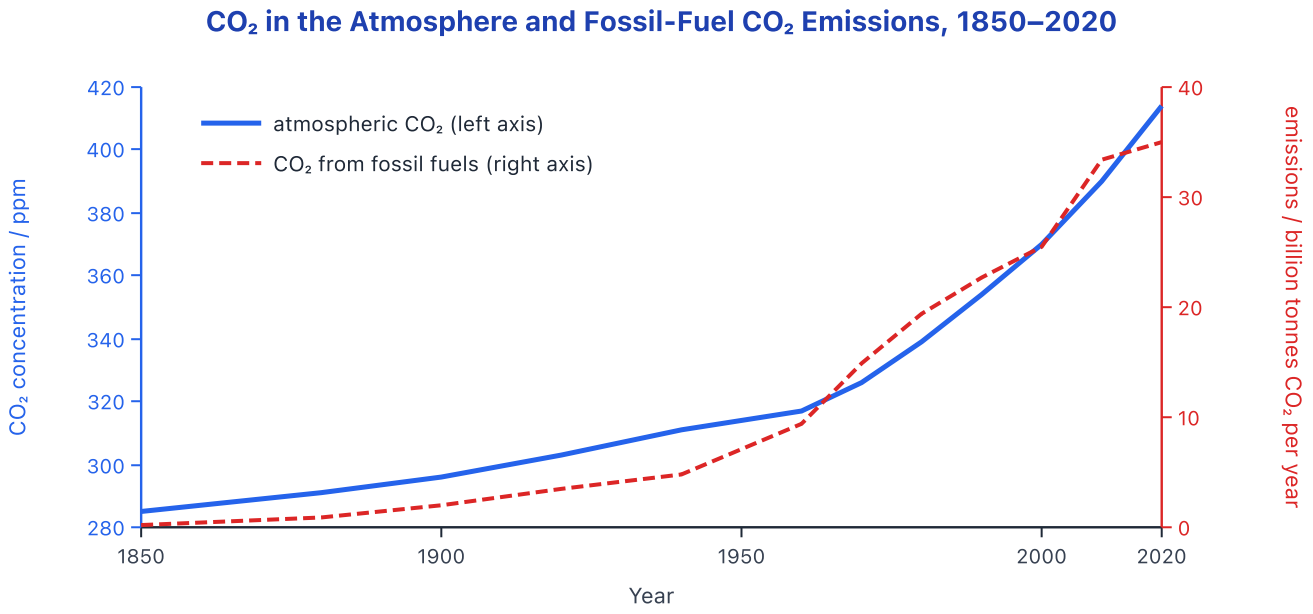
> **Exam Tip:** Greenhouse gases do not "trap heat" or "reflect" radiation. Say they **absorb infrared radiation emitted by the Earth and re-emit it in all directions**. Also, the greenhouse effect is natural and necessary; it is the *enhanced* greenhouse effect from extra greenhouse gases that causes global warming. The ozone layer is a different issue.

Evidence for Human (Anthropogenic) Causes

Over the last 200 years human activities have increased the proportion of greenhouse gases:

- **Carbon dioxide**: burning fossil fuels, deforestation, cement manufacture.
- **Methane**: cattle and other livestock, rice paddy fields, landfill sites, leaks from gas extraction.

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Approximate values (ice-core and Mauna Loa records; global emissions inventories). Both rise together → positive correlation.

- Atmospheric CO_2 has risen from about 280 ppm before the Industrial Revolution to over 410 ppm today. Data come from **air bubbles trapped in ice cores** (past) and direct measurements (since 1958).
- There is a **correlation** between fossil fuel consumption, atmospheric CO_2 concentration and global average temperature (about 1 °C warmer than in 1850).
- There is a plausible **mechanism** (the greenhouse effect) linking the extra CO_2 to warming — so most scientists accept that the correlation reflects **cause and effect**.

Uncertainties in the Evidence

- Past temperatures and CO_2 levels are estimated indirectly (ice cores, tree rings), so they have larger uncertainties; early direct measurements were few and not global.
- The climate is very complex, with many factors (e.g. changes in the Sun's output, volcanic eruptions, ocean currents, cloud cover) affecting temperature.
- Climate **models** are computer models that use patterns in past data and known science to predict the future; different models give a range of predictions. As more data are collected, uncertainty decreases and confidence in the models increases.

> **Ideas about Science (IaS3) — correlation, cause and models.** A correlation alone does not prove one factor causes another; you need a *plausible mechanism*. Individual cases (one hot summer, one cold winter) are not convincing evidence for or against a trend — long-term

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data from many places are needed. Climate science is checked by **peer review**, where other scientists evaluate claims before publication. Physical models (e.g. bottles of air and of CO_2 warmed by a lamp) and computer models are both used to investigate climate change.

Potential Effects of Increased CO_2 and Methane

- **Rising global temperatures** and changes to **where crops can be grown** (some areas too hot or dry; others become suitable); food security issues.
- More frequent and severe **extreme weather** — storms, droughts, heat waves, heavy rainfall.
- **Melting of polar ice** and glaciers, and expansion of warmer sea water → **rising sea levels** → **flooding of low-lying land** and coastal erosion.
- Changes to the distribution of wildlife and spread of disease; more CO_2 dissolving in the oceans makes them more acidic, harming coral and shellfish.

Mitigating the Effects

Method	Scale, risk and environmental implications
Reduce fossil fuel use — renewable and nuclear electricity, energy efficiency, insulation, public transport, electric vehicles	Large investment needed; renewables can be intermittent; nuclear produces radioactive waste; changes in lifestyle needed from billions of people.
Carbon capture and storage (CCS) — capture CO_2 from power stations and store it underground (e.g. in old gas fields)	Expensive; uses energy; only demonstrated on a small scale compared with total emissions; risk of leakage.
Reforestation — trees remove CO_2 by photosynthesis	Needs huge areas of land, which competes with farming; slow; CO_2 released again if forests burn.
Reduce methane — capture landfill gas, change livestock diets, fix gas leaks	Methane is a more powerful greenhouse gas than CO_2 per molecule, so cuts have a big effect.

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Public regulation — international agreements and targets (e.g. net zero), carbon taxes	Needs agreement and enforcement across countries with different economies and priorities.

Because of the very large scale of global emissions, it is difficult to mitigate their effect, and any new measure may have unforeseen environmental impacts.

Ideas about Science (IaS4) — risks, costs, benefits and decisions. Using fossil fuels brings benefits (cheap, reliable energy) but costs and risks (climate change, pollution, unsustainable use of a finite resource). Different countries, groups and individuals may make different decisions about climate change because their *personal, social and economic* circumstances differ — e.g. a developing country may prioritise economic growth, while a low-lying island nation faces flooding. Public regulation (targets for emissions) is needed because individual actions alone are not enough.

Using Data: Orders of Magnitude

You must be able to extract and interpret information from charts, graphs and tables, and use **orders of magnitude** (powers of ten) to judge whether a quantity is significant.

Worked Example: Global fossil-fuel CO_2 emissions are about 3.5×10^{10} tonnes per year. One car emits about 2 tonnes of CO_2 per year. A new low-emission scheme saves 2×10^6 tonnes per year. Is it significant?
Fraction saved = $(2 \times 10^6) \div (3.5 \times 10^{10}) \approx 6 \times 10^{-5}$, i.e. about 0.006% of global emissions — four orders of magnitude smaller. Useful, but on its own it will not have a measurable effect: many such measures are needed.

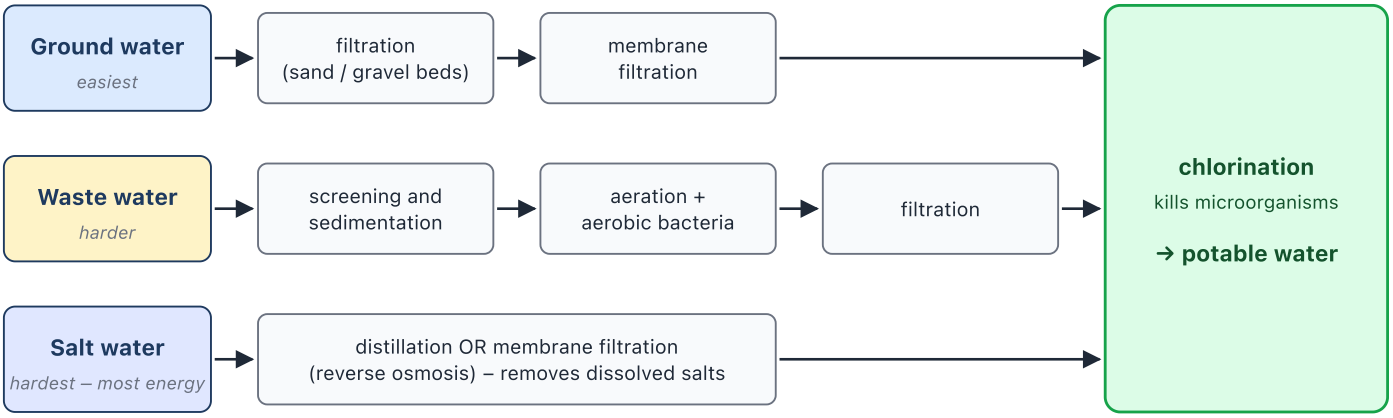
Worked Example: CO_2 makes up about 0.04% of dry air. Express this in parts per million (ppm).
0.04% = 0.04 / 100 = 4×10^{-4}; $\times 10^6$ = **400 ppm**.

C1.4 How Can Scientists Help Improve the Supply of Potable Water?

> **Potable water** is water that is **safe to drink**: it has low levels of dissolved salts and microorganisms. It is *not* chemically pure — it still contains some dissolved substances.

The world's population is increasing, so there is a growing need for potable water. The method used depends on what source is available:

Making Potable Water from Different Sources

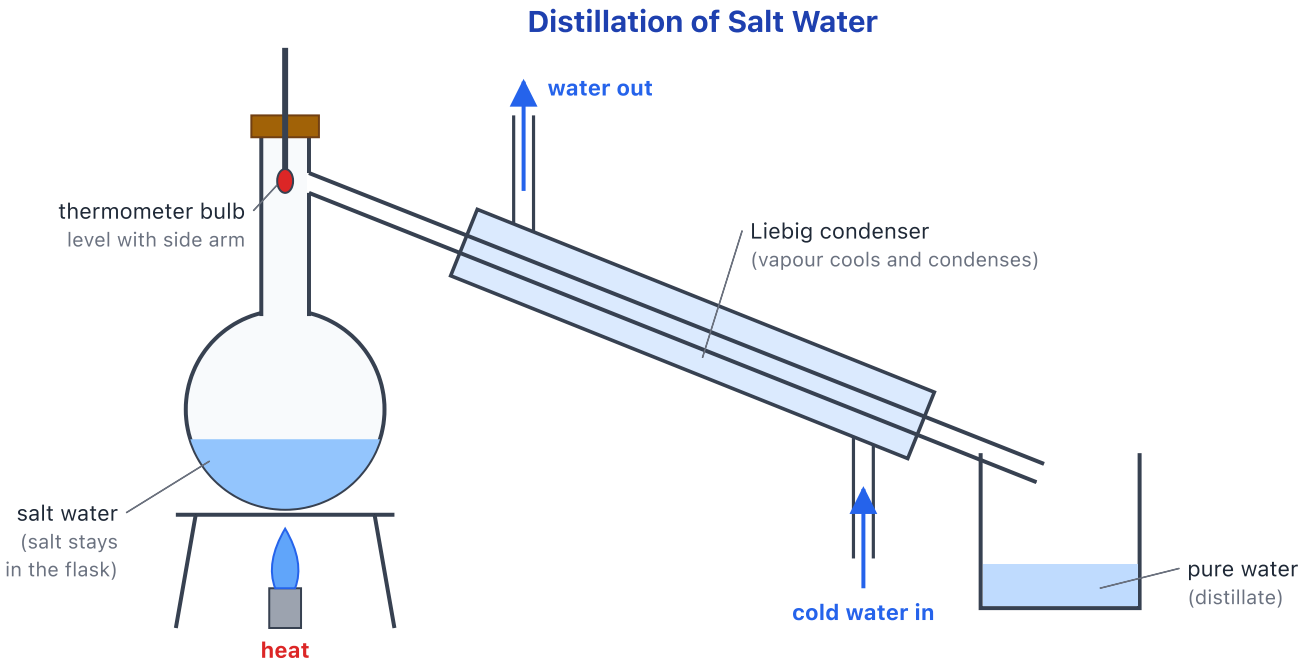


Distilled or reverse-osmosis water from the sea is already free of microorganisms; it may still be chlorinated to protect it in the pipes.

Source	Treatment	Ease / cost
Ground water (from aquifers, rivers, lakes, reservoirs)	**Filtration** through beds of sand and gravel removes solid particles; **membrane filtration** removes very small particles and microorganisms; **chlorination** kills remaining microorganisms	Easiest and cheapest — relatively clean to start with
Waste water (sewage, industrial and agricultural waste)	Screening and sedimentation remove grit and solids; **aeration** — air is bubbled through so **aerobic bacteria** break down organic matter; then **filtration** and **chlorination**	Harder — more stages needed to remove organic matter, harmful microorganisms and chemicals

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Salt water (sea water) — **desalination**	**Distillation** (boil the water and condense the steam, leaving salts behind) or **membrane filtration** (reverse osmosis: water forced through a membrane that holds back dissolved salts)	Hardest and most expensive — needs large amounts of energy; used where fresh water is scarce but energy is available



Exam Tip: On distillation diagrams examiners check two details: the **thermometer bulb is level with the side arm** (it measures the temperature of the vapour going into the condenser), and the **cooling water enters the condenser at the bottom** (lower end) and leaves at the top, so the jacket stays full of cold water.

Chlorination

Chlorine is added to drinking water to **kill microorganisms** and stop the spread of waterborne diseases such as cholera and typhoid. Chlorine is toxic in larger amounts, and can react with organic matter to form small amounts of harmful by-products, but at the low concentrations used the **benefits greatly outweigh the risks**.

Test for Chlorine

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Gas	Test	Positive result
Chlorine, Cl_2	Hold **damp blue litmus paper** in the gas	Turns **red** (chlorine is acidic) and then is **bleached white**

Chlorine is toxic — carry out the test in a fume cupboard. The practical "identify unknown gases" (PAG2) brings together the tests for oxygen, hydrogen, carbon dioxide and chlorine.

> **Ideas about Science (IaS4).** Water treatment technologies make a big *positive difference* to people's lives by preventing disease. Access to treated water raises issues of **risk, cost and benefit**: in some countries chlorination is required by public regulation, but other parts of the world still lack treated water, leading to a higher risk of disease. Providing treated water for everyone raises **ethical** questions — who should pay, and is it right that some people have safe water while others do not? A common ethical argument is that the right decision gives the best outcome for the greatest number of people.

Summary & Key Equations

- Particle model: arrangement, movement and attractions explain states; particles do not have bulk properties
- (HT) Particle diagrams are limited: wrong scale, inelastic spheres, forces between particles not fully shown
- Early atmosphere: volcanic CO_2 + H_2O; oceans formed by condensation; CO_2 removed by oceans, rocks, photosynthesis; O_2 built up
- CO and particulates ← incomplete combustion; SO_2 ← sulfur impurities; NO_x ← N_2 + O_2 at high temperature
- $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ · $2CO + 2NO \rightarrow 2CO_2 + N_2$ · $2H_2 + O_2 \rightarrow 2H_2O$
- Oxidation = gain of oxygen; combustion is oxidation
- Exothermic: surroundings warm up; endothermic: surroundings cool down; activation energy = minimum energy needed to react
- (HT) Energy change = bonds broken − bonds made (negative = exothermic)

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- Greenhouse gases absorb and re-emit infrared; extra CO_2 and CH_4 from human activity correlate with rising temperatures
- Potable water: filtration, membrane filtration, aeration with bacteria, chlorination, distillation of salt water
- Gas tests: O_2 relights glowing splint; H_2 squeaky pop; CO_2 limewater milky; Cl_2 bleaches damp blue litmus (red first)

> **Exam Tip:** When asked to "evaluate" a method (fuel cells, carbon capture, desalination), give advantages **and** disadvantages, then finish with a **judgement** that refers back to the context given in the question.

Specification Checklist

Ref	Can you...
C1.1.1	explain the particle model for states and changes of state, tell physical from chemical changes, and explain that particles do not have the bulk properties of the substance
C1.1.2 (HT)	explain the limitations of the particle model for changes of state when particles are drawn as inelastic spheres
C1.1.3	use energy transfers and the strength of forces between particles to explain different melting and boiling temperatures
C1.1.4	use melting and boiling point data to predict states
C1.1.5	interpret evidence for how the early atmosphere formed
C1.1.6	describe how an oxygen-rich atmosphere developed
C1.1.7	describe the sources of CO and particulates, SO_2 and nitrogen oxides
C1.1.8	explain the problems these pollutants cause and how catalytic converters, low-sulfur petrol and gas scrubbers reduce emissions

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C1.1.9	write formulae of elements and simple covalent compounds
C1.1.10	use conservation of mass to write formulae and balanced equations
C1.1.11	use arithmetic and ratios to balance equations
C1.1.12	describe the tests for oxygen, hydrogen and carbon dioxide (PAG2)
C1.1.13	explain oxidation as gain of oxygen
C1.2.1	distinguish exothermic and endothermic reactions by the temperature change of the surroundings
C1.2.2	draw and label reaction profiles, showing the activation energy
C1.2.3	explain activation energy as the energy needed for a reaction to occur
C1.2.4	interpret charts and graphs of reaction profiles
C1.2.5 (HT)	calculate energy changes from bond-breaking and bond-making energies
C1.2.6	carry out the arithmetic for energy-change calculations
C1.2.7	describe an investigation to decide whether a reaction is exothermic or endothermic
C1.2.8	recall that a chemical cell produces a potential difference until the reactants are used up
C1.2.9	evaluate hydrogen–oxygen and other fuel cells for given uses
C1.3.1	describe the greenhouse effect in terms of radiation interacting with matter
C1.3.2	evaluate evidence for human causes of climate change, including the CO_2–fossil fuel correlation, and describe the uncertainties
C1.3.3	describe the potential effects of more CO_2 and methane: crops, extreme weather, melting ice, flooding

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C1.3.4	describe how the effects can be mitigated, considering scale, risk and environmental implications
C1.3.5	extract and interpret information from charts, graphs and tables
C1.3.6	use orders of magnitude to judge the significance of data
C1.4.1	describe how ground, waste and salt water are made potable: filtration, membrane filtration, aeration, bacteria, chlorination, distillation
C1.4.2	describe the test for chlorine using damp blue litmus paper (PAG2)

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